Recall: Components in a graph G are equivalence classes under the relation on V(G) defined by $u \mathcal{R} v$ iff there is a u–v path.

Definition. *Let* G *be a graph. We define a relation* $\mathcal{S}$ *on* E(G) *by* $e \mathcal{S} f$ *iff* $e = f$ *or there is a cycle on* G *that uses* e *and* f.

Theorem (5.8). *The relation* $\mathcal{S}$ *defined above is an equivalence relation.*

Proof:

1) Reflexive: $e \mathcal{S} e$ ✓

2) Symmetric: $e \mathcal{S} f \Rightarrow f \mathcal{S} e$ ✓

3) Transitive: $e \mathcal{S} f$, $f \mathcal{S} g \Rightarrow e \mathcal{S} g$

Assume - e, f lie on a cycle C
- f, g lie on a cycle C'

WTS: There is a cycle C'' containing e, g.

Fix some $e \in E(G)$

Class of $e = \{ f \in E(G)$ such that $e \mathcal{S} f \}$

1) Case: g is in C or e in C' ✓

2) Case: g not in C and e not in C':

Let $P = C - e$

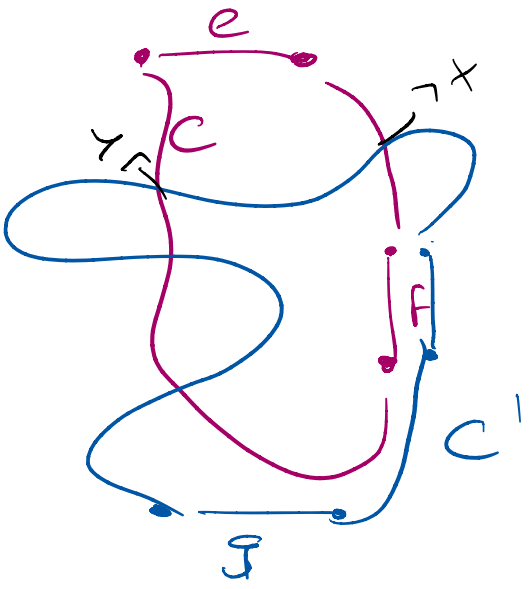
Let x = first vertex in $P \cap C'$
y = last vertex in $P \cap C'$ } Exist?

Let P' = x-y path in C' using g

P'' = y-x path in C using e

Take C'' to be P' followed by P'' ✓

is a cycle? using e, g?

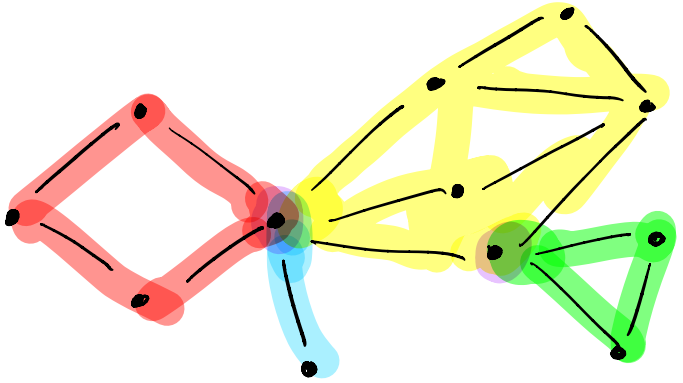


Last time:

- $v \in V(G)$ is a cut-vertex if $G - v$ has more components than G.
- We say that a connected graph G is non-separable if it contains no cut-vertices.
- G is non-separable iff every two vertices lie on a common cycle.

Definition. *A **block** of G is a maximal non-separable subgraph of G.*

Example. *Identify cut-vertices and blocks in the following graph*



Alternative definition of blocks:

- Define: equivalence relation on $E(G)$: $e \sim f$ iff $e = f$ or there is a cycle on G that uses e and f.
- Let $\mathcal{E}_1, \ldots, \mathcal{E}_k$ denote the equivalence classes under $\sim$. Thus $E(G) = \mathcal{E}_1 \sqcup \cdots \sqcup \mathcal{E}_k$.
- Each equivalence class $\mathcal{E}_i$ induces a subgraph B_i with $E(B_i) = \mathcal{E}_i$ and $V(B) = \bigcup_{e \in \mathcal{E}_i} e$ (all vertices of G incident to some edge in $\mathcal{E}_i$.)

Theorem. *Each B_i is a block and all blocks of G arise in this way.*

Proof:

1) Each B_i is a block $\Leftarrow$ non-separable ✓
$\Rightarrow$ if we add an edge, it becomes separable.

On B_i there are no cut-vertices (any two edges on B_i lie on some common cycle). ✓

If we add an edge to B_i, it becomes separable (edges outside B_i lie on a different class).

2) Any block arises like This:

Let B be a block. Let us show: any two edges in B are related under S (lie in a common cycle).

Exercise (5.10): Let G connected graph with $|V(G)| \geq 2$.

Then G is non-separable $\Leftrightarrow$ any two adjacent edges lie on a common cycle.

Given edges $e = uv$, $f = xy$ edges in B

There is a v-x path in B (connected)

$v = w_0, w_1, w_2, \ldots, w_k = x$

$e_0 \; S \; e_1 \; S \; e_2 \; S \ldots S \; e_k$

$e_0 S e_1$, $e_1 S e_2 \ldots$, $e_k S f$

B is non separable

Corollary (5.9). *Every two distinct blocks* B_1 *and* B_2 *in a connected graph* G *satisfy:*

(a) B_1 *and* B_2 *are edge-disjoint.*

(b) B_1 *and* B_2 *have at most one common vertex.*

(c) If B_1 *and* B_2 *have a vertex* v *in common, then* v *is a cut-vertex of* G.

Proof

a) For free, $E(G) = E_1 \sqcup \dots \sqcup E_k$ disjoint union

↳ Edges of B_i

b)

Assume $u \neq v$ are both in B_1 and B_2

Let P_1 a u-v path in B_1

P_2 a u-v path in B_2

Want a contradiction: There is a cycle in G using an edge on B_1 and an edge on B_2

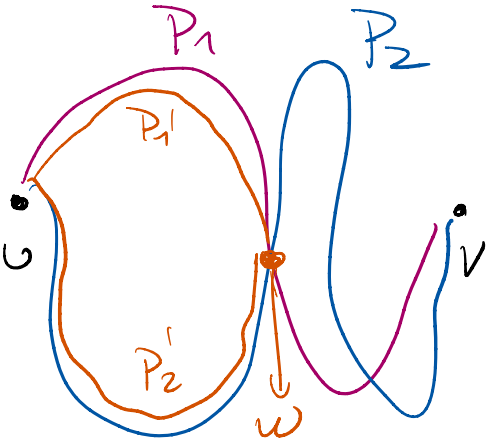
Note: P_1 and P_2 are edge-disjoint (a)

Let w first vertex on $P_1 \cap P_2$ after v

w will exist because $v \in P_1 \cap P_2$.

Let $P_1' = v$-w path in P_1

$P_2' = w$-v path in P_2



Then P_1' followed by P_2' form a cycle in G using an edge from B_1 and an edge from B_2, contradiction!

(edges on B_1 are <u>not</u> related to edges on B_2).

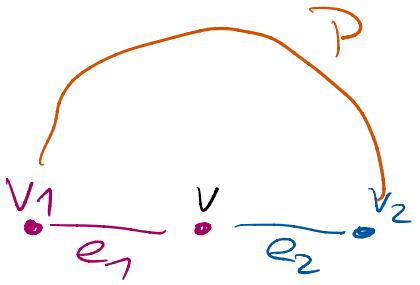
under the relation S

c) Assume B_1 and B_2 intersect at a vertex v.

WTS: v is a cut vertex.

We have edges $e_1 = v_1 v$ in B_1

$e_2 = v v_2$ in B_2



If v is <u>not</u> a cut vertex, There is some v_1-v_2 path in G <u>not</u> using v

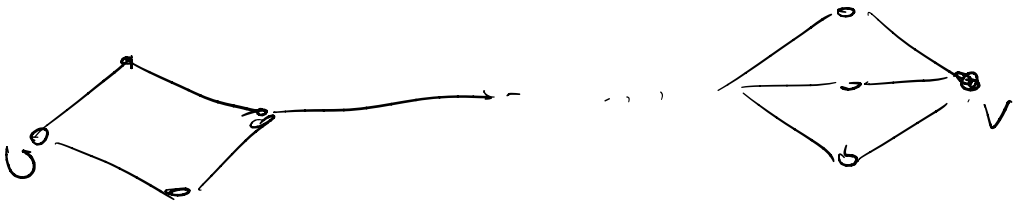
Then P, e_1, e_2 form a cycle on G using an edge on B_1 (e_1) and an edge on B_2 (e_2)

Contradiction: edges on B_1 are <u>not</u> related ✓ under S to edges on B_2

∎

Exercise (5.6)**.** *Prove that a* 3*-regular graph* G *has a cut-vertex if and only if* G *has a bridge.*

Exercise (5.13)**.** *Assume* G *is a connected graph with cut-vertices. Prove or disprove: if* $u, v \in V(G)$ *are such that* $d(u, v) = diam G$*, then no block of* G *contains* u *and* v*.*



Definition. *Let* G *be a graph* (connected).

- *A* ***vertex-cut*** *is a subset* $U \subset V(G)$ *such that* $G - U$ *is disconnected.* (more components than G)
- *The* ***vertex-connectivity*** $\kappa(G)$ *is the minimum cardinality of a vertex-cut. For* $G = K_n$, *we set* $\kappa(G) = n-1$.
- *If* k *is a non-negative integer, we say that* G *is* k*-connected if* $\kappa(G) \geqslant k$.

Note: : Assume G has n vertices.

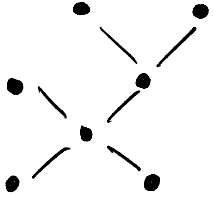
(1) $0 \leqslant \kappa(G) \leqslant n-1$

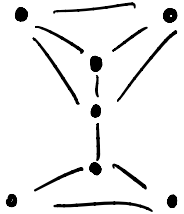
(2) $\kappa(G) = 0$ iff G is disconnected

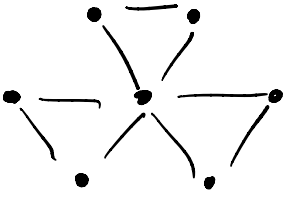
(3) $\kappa(G) = 1$ iff G has a cut-vertex and G is connected

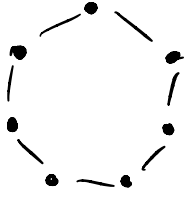
(4) G is 2-connected iff G is non-separable and $|V(G)| \geq 2$

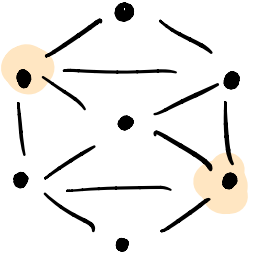
Example. *Compute* $\kappa(G)$ *for the following graphs.*

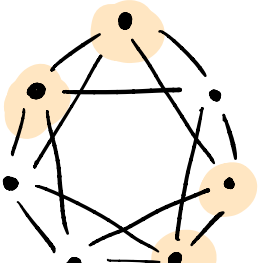












Definition. *Let* G *be a graph*

- *An* ***edge-cut*** *is a subset* $F \subset E(G)$ *such that* $G - F$ *is disconnected.*
- *The* ***edge-connectivity*** $\lambda(G)$ *is the minimum cardinality of an edge-cut. For* $G = K_1$*, we set* $\lambda(G) = 0$*.*
- *If* k *is a non-negative integer, we say that* G *is* k*-edge-connected if* $\lambda(G) \geqslant k$*.*

Note: : Assume G has n vertices.

(1) $0 \leqslant \lambda(G) \leqslant n - 1$ (2) $\lambda(G) = 0$ iff G disconnected (3) $\lambda(G) = 1$ iff G has a bridge.

Example. *Compute* $\lambda(G)$ *for the following graphs.*

